Answer Key & Teacher Notes

Session 1 — Distance Between Points on a Grid · Session 2 — Comparing Distances on the Plane

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Distance Between Points on a Grid

TALK PROMPTS

- How can the coordinates alone tell you whether a trip runs across or up and down?
- Why does a trip that crosses the x-axis join the two distances instead of taking one away?

PARTNER TASK

Form A, problem 1. One partner finds the distance from (3, 14) to (3, 4) (part a); the other finds the distance from (−11, 2) to (6, 2) (part b). Each explains whether they added or subtracted, then they switch and check.

WATCH FOR

- Subtracting the absolute values when the points are on opposite sides of an axis ((2, −9) to (2, 6) as 3 instead of 15).
- Using the coordinate that stays the same instead of the one that changes.
- Changing the y-coordinate for a move left or right when finding a missing point.

SESSION 2 · TEACHER NOTES

Comparing Distances on the Plane

TALK PROMPTS

- How do the signs of the changing coordinates tell you whether to add or to subtract?
- How can the coordinates alone tell you which ride crosses an axis?
- Why must both distances use the same scale before you compare them?

PARTNER TASK

Form A, problem 1. Both trips start at (5, −3). One partner finds the distance to (5, −15) (part a); the other finds the distance to (5, 10) (part b). Compare: who added, who subtracted, and why? Then decide which trip is longer.

WATCH FOR

- Subtracting absolute values for points on opposite sides of an axis ((−8, 3) to (4, 3) as 4 instead of 12).
- Comparing grid units for one trip with scaled meters for the other trip.
- Writing a negative distance, such as −9, instead of a length.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.8	1	2	1	1	2	2	2	3	2	1	1
S1 · Form B	6.NS.C.8	1	2	1	1	2	2	2	3	2	1	1
S2 · Form A	6.NS.C.8	1	2	2	2	2	2	2	3	2	1	1
S2 · Form B	6.NS.C.8	1	2	2	2	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

distance: How far apart two points are. It is never **negative**.

absolute value: How far a number is from **0**, ignoring its sign.

horizontal distance: Distance side to side. Use the **x**-coordinates.

vertical distance: Distance up and down. Use the **y**-coordinates.

- If two points share a y-coordinate, find the distance with the **x**-coordinates.
- If the changing coordinates have opposite signs, **add** their absolute values. If they have the same sign, **subtract** them.

Example: The points share the y-coordinate **3**, so use the x-coordinates. −4 and 5 have **opposite** signs, so the trip crosses the y-axis. Add the absolute values: 4 + 5 = **9**. Answer: The distance is **9** units.

YOU TRY

1. a. 6 + 3 = 9 units b. 7 − 2 = 5 units

2. B = (3, 4)

EXIT TICKET

1. a. 3 + 6 = 9 units b. 8 − 2 = 6 units

2. B = (−2, −2)

3. **B** (horizontal line)

PRACTICE FORM A

1. a. 14 − 4 = 10 b. 11 + 6 = 17 c. 9 + 6 = 15 d. 6 + 5 = 11

2. a. L(−8, 5), P(7, 5), C(−8, −6) b. same y; 8 + 7 = 15 km c. same x; 5 + 6 = 11 km

3. 13 + 4 = 17 m

4. 20 − 7 = 13 units

5. H = (−5, −3)

6. 18 − 7 = 11 ft. Both y-coordinates are positive (same side of the x-axis), so subtract.

7. the x-coordinates; 17 + 12 = 29 blocks

8. a. 4 + 5 = 9 blocks b. 6 + 3 = 9 blocks c. 9 + 9 = 18 blocks

9. The y-coordinates −9 and 6 have opposite signs, so the trip crosses the x-axis. Malik should add: 9 + 6 = 15 units.

10. **B** (7 units)

11. **B** (9 units)

Self-check word **LENGTH**

PRACTICE FORM B

1. a. 12 − 5 = 7 b. 9 + 7 = 16 c. 8 + 5 = 13 d. 5 + 9 = 14

2. a. M(−6, 4), U(8, 4), S(−6, −7) b. same y; 6 + 8 = 14 km c. same x; 4 + 7 = 11 km

3. 11 + 5 = 16 m

4. 20 − 8 = 12 units

5. H = (−5, −2)

6. 17 − 8 = 9 ft. Both y-coordinates are positive (same side of the x-axis), so subtract.

7. the x-coordinates; 15 + 13 = 28 blocks

8. a. 5 + 6 = 11 blocks b. 4 + 2 = 6 blocks c. 11 + 6 = 17 blocks

9. The y-coordinates −8 and 5 have opposite signs, so the trip crosses the x-axis. Amina should add: 8 + 5 = 13 units.

10. **B** (9 units)

11. **B** (vertical line)

Self-check word **ORIGIN**

GUIDED NOTES

quadrant: One of the four **sections** of the coordinate plane.

compare: Decide which amount is **greater** or less.

scale: How much real length one grid **unit** stands for.

- Find **each** distance first. Then compare the two numbers.
- With a scale, change **both** distances to real units before you compare.

Example: A to B: same x, so 9 − 2 = **7** units = 35 m. A to C: same y, opposite signs, so 1 + 8 = **9** units = **45** m. Answer: **C** is farther, by 10 m.

YOU TRY

1. a. add: 2 + 7 = 9 units b. subtract: 12 − 5 = 7 units

2. Ana: 3 + 6 = 9 units; Ben: 4 + 6 = 10 units. Ben walks farther, by 1 unit.

EXIT TICKET

1. a. 5 + 4 = 9 units b. 9 − 1 = 8 units

2. Lee: 6 + 3 = 9 units. Mia: 4 + 7 = 11 units. Mia, by 2 units.

3. **A** (5 units)

PRACTICE FORM A

1. a. 15 − 3 = 12 b. 3 + 10 = 13 c. 4 + 6 = 10 d. 8 − 2 = 6

2. Gym(−6, 4), Field(7, 4), Office(−6, −5). a. 6 + 7 = 13 units = 130 m b. 4 + 5 = 9 units = 90 m c. Gym to Field, by 40 m

3. Tariq: 5 + 8 = 13 blocks. Jada: 9 + 6 = 15 blocks. Jada, by 2 blocks.

4. U = (−4, −6)

5. M = (−7, −5)

6. Kendra: 12 − 3 = 9 km. Owen: 2 + 10 = 12 km. Owen, by 3 km.

7. Ana: 5.75 + 3.25 = 9 units. Ben: 9.75 − 5.75 = 4 units. Ana, by 5 units.

8. a. Stop 1: 12, Stop 2: 9, Stop 3: 4 units b. Stop 3, Stop 2, Stop 1 c. 12 + 9 = 21 units

9. The x-coordinates −8 and 4 have opposite signs, so the points are on opposite sides of the y-axis. Ines should add: 8 + 4 = 12 units.

10. **B** (subtraction)

11. **C** (negative)

Self-check word **LONGER**

PRACTICE FORM B

1. a. 13 − 2 = 11 b. 2 + 8 = 10 c. 3 + 6 = 9 d. 9 − 1 = 8

2. Pond(−7, 5), Meadow(5, 5), Cabin(−7, −2). a. 7 + 5 = 12 units = 120 m b. 5 + 2 = 7 units = 70 m c. Pond to Meadow, by 50 m

3. Rosa: 4 + 7 = 11 blocks. Kofi: 8 + 6 = 14 blocks. Kofi, by 3 blocks.

4. U = (−3, −6)

5. M = (−6, −4)

6. Mei: 13 − 4 = 9 km. Andre: 3 + 11 = 14 km. Andre, by 5 km.

7. Ana: 4.75 + 2.25 = 7 units. Ben: 9.75 − 4.75 = 5 units. Ana, by 2 units.

8. a. Stop 1: 11, Stop 2: 8, Stop 3: 5 units b. Stop 3, Stop 2, Stop 1 c. 11 + 8 = 19 units

9. The x-coordinates −6 and 9 have opposite signs, so the points are on opposite sides of the y-axis. Diego should add: 6 + 9 = 15 units.

10. **B** (addition)

11. **C** (14 units)

Self-check word **METERS**

SHEET 1 · DISTANCE BETWEEN POINTS ON A GRID

1. A(−4, 3), B(3, 3), C(3, −2), D(−4, −4). a. 4 + 3 = 7 units b. 3 + 2 = 5 units c. 3 + 4 = 7 units
2. a. 8 b. 5 c. 9 d. 11 e. 5 f. 6.5 g. 10 h. 4.5
3. Same x; 3 + 5 = 8 blocks
4. Q = (−4, −3)
5. Both y-coordinates are negative (same side of the x-axis), so subtract: 6 − 2 = 4 units.

SHEET 2 · COMPARING DISTANCES ON THE PLANE

1. H(−4, 2), L(3, 2), P(−4, −3). a. 4 + 3 = 7 blocks b. 2 + 3 = 5 blocks c. the Park, by 2 blocks
2. a. 8 − 3 = 5 b. 6 + 7 = 13 c. 7 − 1 = 6 d. 2.5 + 4.5 = 7
3. Ana: 5 + 6 = 11 units. Jay: 4 + 6 = 10 units. Ana runs 1 unit farther.
4. Add. The y-coordinates 7 and −2 have different signs, so the path crosses the x-axis: 7 + 2 = 9 units.
5. **A** (JK, by 2 units)
6. **C** (8 units)

October 2026 conceptual additions — teacher key

Session 1 · Distance Between Points on a Grid

Shared y means horizontal segments. Distances 7 and 2 units; the first crosses the y-axis, the second stays on one side.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Shared y means horizontal segments. Distances 7 and 2 units; the first crosses the y-axis, the second stays on one side.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Comparing Distances on the Plane

AB 6 units=18 m; AC 8 units=24 m. AC is 6 m longer.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: AB 6 units=18 m; AC 8 units=24 m. AC is 6 m longer.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.